

MID EXAMINATION – SPRING 2026Computer Architecture

CSE:UG1(PC)

Duration: 90 Mins (2.30 PM to 4.00 PM)

Date: 24 Feb 2026

Max. Marks: 30

Roll No: _____

Section: _____

Instructions:

1. All questions are compulsory.
2. Write the answers to the point and legibly..
3. All sub-parts of a question should be written together.
4. Attach the question paper with the answer sheet.

Part A [9 Marks]

- 1) The binary number 101.11100_2 is to be rounded to 3 fractional bits using the round to nearest even (RNE) rule. What is the correctly rounded result?
a) 101.110 b) 101.111 **c) 110.000** d) 101.100
- 2) For the following declaration, which of the given options are NOT always true?
`int x = ...;   float f = ...;   double d = ...;`
a) `x == (int)(float) x` b) `x == (int)(double) x` c) `f == -(-f);` **d) `(d+f)-d == f`**
- 3) What does this expression evaluate to ? `~ ( ~a | (b ^ (MIN_INT + MAX_INT)))`
a) `~(a^b)` **b) `a&b`**
c) `~(a+b)` d) `a^b`
- 4) What gets printed for “`printf("%d\n", -1 < 0U);`”
a) `0` b) `1`
c) Compilation error d) Undefined behaviour
- 5) If an 8-bit signed number 0xF2 is converted to 32-bit signed, what is the new hex value?
a) 0x000000F2 b) **0xFFFFFFF2**
c) 0x0000000F d) 0xFFFFFFFF
- 6) A _____ is the operating system’s abstraction for a running program.
a) RAM b) ROM **c) Process** d) Cache
- 7) Multiple processes can run concurrently on the same system, and each process appears to have exclusive use of the hardware. The operating system performs this interleaving with a mechanism known as _____.
a) Context Switching b) Interface c) Abstraction d) All of the above
- 8) An application program to manipulate the hardware must go through the _____.
a) Process b) Program **c) Operating System** d) None of the above
- 9) What is the output for the following code

```
short s = -2;
unsigned u = s;
printf("%u\n", u);
```

1111111111111111 11111111 11111110

Or $2^{32}-2$

Part B.

1.

- a. Multiply the following numbers given in IEEE 754 single-precision format and give the solution in decimal. [5 Marks]

Num1: 1 10000011 100110000000000000000000

Num2: 0 10000000 110000000000000000000000

1. (Marking Scheme proper decoding num1 2 marks
2. decoding num 2 ---- 2 marks
3. multiplying ----- 1 mark

IEEE-754 (single precision):

- 1 sign bit
- 8 exponent bits (bias = 127)
- 23 fraction bits
- Implicit leading 1 for normalized numbers

Decode Num1

Sign

1 → negative

Exponent

$10000011_2 = 131_{10}$

Unbiased exponent:

$131 - 127 = 4$

Mantissa

Add hidden 1:

1.100110000000000000000000

Convert to decimal:

$1.10011 = 1.59375$

Num1 = $-(1.59375 \times 2^4)$
 = $-(1.59375 \times 16) = -25.5$

Decode Num2

Sign

0 → positive

Exponent

$10000000_2 = 128_{10}$

Unbiased exponent:

$$128 - 127 = 1$$

Mantissa

$$1.110000000000000000000000$$

$$= 1 + 0.5 + 0.25 = 1.75$$

$$\text{Num}_2 = 1.75 \times 2^1$$

$$= 1.75 \times 2$$

$$= 3.5$$

Multiply Mantissas

$$1.59375 \times 1.75$$

$$= 2.7890625$$

$$\text{So the product is: } 2.7890625 \times 2^5$$

Normalize

Since $2.7890625 \geq 2$, shift right by 1:

$$= 2.7890625 = 1.39453125 \times 2$$

Exponent increases by 1:

$$= 1.39453125 \times 2^6$$

$$= -89.25$$

- b. Fill in the following table. Provide the corresponding integer value of the 5-bit arguments, the value of both their integer and two's complement sums, the bit-level representation of the two's complement sum, and the type of overflow. The first row is filled as an example

[2Marks]. [Marking each correct answer 0.2×10 ----- 2 marks]

a	b	a+b	$a +_5^t b$	Overflow
10101 (-11)	10011 (-13)	101000 (-24)	01000 (8)	Negative
11001 (-7)	11001 (-7)	110010 (-14)	10010 (-14)	Normal
10110 (-10)	01001 (9)	111111 (-1)	11111 (-1)	Normal

2. Suppose you work as a truck driver, and you have been hired to carry a load of potatoes from Boise, Idaho, to Minneapolis, Minnesota, a total distance of 2,500 kilometers. You estimate you can average 100 km/hr driving within the speed limits, requiring a total of 25 hours for the trip.

- a. You hear on the news that Montana has just abolished its speed limit, which constitutes 1,500 km of the trip. Your truck can travel at 150 km/hr. What will be your speedup for the trip?

[1 Mark]

Marking Scheme right answer ----- 1 otherwise 0 no partial marking

his will give a speedup of $25/(10 + 10) = 1.25\times$.

- b. You can buy a new turbocharger for your truck at www.fasttrucks.com. They stock a variety of models, but the faster you want to go, the more it will cost. How fast must you travel through Montana to get an overall speedup for your trip of $1.67\times$?

Marking Scheme right answer ---- 3 marks with proper justification otherwise 0 no partial marking

Total distance = 2500 km
 Original speed = 100 km/hr
 Original time = $2500 / 100 = 25$ hours
 Required speedup = $1.67\times$

Step 1: Calculate new total time
 New time = $25 / 1.67 \approx 15$ hours
 Step 2: Time for non-Montana portion
 Distance outside Montana = 1000 km
 Speed = 100 km/hr
 Time = $1000 / 100 = 10$ hours
 Step 3: Time available for Montana portion
 Time available = $15 - 10 = 5$ hours
 Step 4: Required speed in Montana
 Distance in Montana = 1500 km
 Required speed = $1500 / 5 = 300$ km/hr
 You must travel at 300 km/hr through Montana to achieve an overall speedup of $1.67\times$.

3. A car manufacturing company has promised their customers that the next release of a new engine will show a $4\times$ performance improvement. You have been assigned the task of delivering on that promise. You have determined that only 90% of the engine can be improved. How much (i.e., what value of k) would you need to improve this part to meet the overall performance target of the engine?

[3

Marks]

Marking Scheme right answer ---- 3 marks with proper justification otherwise 1 mark if amdahls law is used but no correct answer

Required overall speedup = $4\times$
Fraction that can be improved (f) = 0.9
Fraction that cannot be improved = 0.1

Using Amdahl's Law:
Speedup = $1 / [(1 - f) + (f / k)]$

Substituting values:
 $4 = 1 / [0.1 + (0.9 / k)]$

Taking reciprocal:
 $0.1 + (0.9 / k) = 0.25$

Solving:

$$0.9 / k = 0.25 - 0.1$$

$$0.9 / k = 0.15$$

$$k = 0.9 / 0.15$$

$$k = 6$$

The improved 90% portion of the engine must be made 6× faster to achieve an overall 4× performance improvement.

4.

a. Consider the following C code executed on a 32-bit two's complement system **[3Marks]**

short s = -1;

unsigned u1 = s;

unsigned u2 = (unsigned short)s;

i. What is the difference between the two assignments.

Marking Scheme right answer — 2 marks with proper justification of case 1 and case 2 and one mark for second part if answer is right otherwise 0

Representation of s

On a 32-bit two's complement system:

- short = 16 bits
 - unsigned = 32 bits
- short s = -1;

In 16-bit two's complement:

$$-1 = 1111\ 1111\ 1111\ 1111 = 0xFFFF$$

Case 1:

unsigned u1 = s;

Here: s (16-bit signed) is first promoted to 32-bit signed int

Then converted to 32-bit unsigned

Since s = -1, when promoted to 32-bit:

11111111 11111111 11111111 11111111 (32-bit signed -1)

Now converting to unsigned does not change bits, only interpretation.

So:

$$u1 = 2^{32} - 1 = 4294967295$$

Case 2: unsigned u2 = (unsigned short)s;

Here:

First convert s to unsigned short (16-bit)

Then assign to 32-bit unsigned

Since:

(unsigned short)-1 = 0xFFFF = 65535

Now zero-extended to 32 bits:

00000000 00000000 11111111 11111111

So:

u2=65535

ii. Determine the values stored in u1 and u2.

u1 : 4294967295 (Sign-extended to 32 bits before unsigned conversion)

u2:65535 (Converted to unsigned 16-bit first, then zero-extended)

- b. Why does the two's complement system not have a positive counterpart for the most negative number? discuss the effect of negating the smallest representable number. [2Marks]

Marking Scheme right answer ---- 2 marks with proper justification otherwise 0

For an n-bit two's complement system:

Range= -2^{n-1} to $2^{(n-1)}-1$ For 8-bit: -128 to +127

The smallest number is:

-2^{n-1}

There is no $+2^{n-1}$, because one value is used to represent zero.

Effect of Negating the Smallest Number

Example (8-bit):

$x=-128=10000000(\text{base}2)$

Negating:X

Invert bits $\rightarrow 01111111$

Add 1 $\rightarrow 10000000$

We get the same number again (-128).

This causes overflow, because +128 cannot be represented in 8-bit two's complement.